METAL LEDGER & VALUATION ENGINE

System Overview & Testing Guide

What was built, why, and how to verify it — in plain language, with the task brief's own numbers.

Prepared October 2026 · Based on Task Brief dated Oct 13, 2026

What this module is, in one paragraph

This is a backend module for the Gold & Silver Shop ERP — it owns every weight, rate, valuation and metal balance the shop relies on. It has no screens of its own; it's accepted on its rules, not its UI. Four tasks were built this week: converting between weight units and purities, keeping one authoritative rate book, recording every movement of metal in or out of the shop, and pricing buy-backs. Two further tasks — karigar reconciliation and zakat reporting — are deliberately left for later, with the design already accounting for them.

30-second version, if that's all the time there is

1. **Weights:** converts gram ↔ tola/masha/ratti/point without ever losing precision.
2. **Rates:** one official price per metal; the owner's manual correction always beats an automatic update.
3. **Ledger:** every movement of metal is recorded permanently; mistakes are corrected by adding a new entry, never by erasing the old one.
4. **Buy-back:** always priced fresh, today, off the re-weighed grams and today's rate — the original sale price can't reach it, by design.

Task 1 — One language for weight and purity

> FR-1, G-4
> *"Any weight the shop speaks must mean exactly one thing… A weight entered in any unit and read back in another returns the same quantity, both directions, every time… Purity turns a worn weight into pure-metal weight: a 10 g 22K chain is 9.167 g of pure gold, a 10 g 925 bar is 9.25 g of pure silver."*

What it does: the shop uses two weight systems — grams and the old tola/masha/ratti/point system. This converts between them without drift, and works out how much pure gold or silver is actually inside an item based on its purity.

Test 1 — a mixed-unit gold item

INPUT	VALUE
Weight	50.5 grams
Purity	22K gold (91.66%, owner-confirmed)

RESULT

In traditional units: **4 tola, 3 masha, 7 ratti, 64.49 point** — converts back to 50.5g, within floating-point noise. A 10g 22K item contains **9.166g** of pure gold.

Test 2 — the brief's own example, plus a silver check

INPUT	VALUE
Weight	1 tola gold (11.6638038g)
Purity	22K (91.66%)

RESULT

Pure gold content: **10.691g**. And the brief's own silver example, confirmed exactly: 10g at 925 (92.50%) = **9.25g pure**.

Task 2 — The rate book

> **FR-2, FR-3, FR-4, G-2**
>
> *"Every price in the system must trace back to one rate... Nothing is ever overwritten or deleted... An Admin rate outranks a fetched rate for the same metal, purity and day... a 9 a.m. fetch, an 11 a.m. correction and a 1 p.m. retry [must not] silently erase the owner's decision."*

What it does: every price the shop uses comes from one official place. Prices arrive automatically or are typed in by the owner — and the owner's price always wins, even against a later automatic update the same day.

Test 1 — the brief's exact 9am / 11am / 1pm scenario

TIME	ACTION	RATE	BECOMES CURRENT?
9:00 AM	Auto-fetch	Rs 22,000/g	Yes — until…
11:00 AM	Admin correction	Rs 22,400/g	Yes
1:00 PM	Auto-fetch again	Rs 22,150/g	**No — rejected**

RESULT

3 entries kept in history, **Rs 22,400/g stays in force** — matching the brief's "Done when" line word for word.

Test 2 — a purity with no rate of its own (derived)

INPUT	VALUE
24K gold's current rate	Rs 34,000/g
21K gold	has no rate entered today

RESULT

Derived 21K rate: **Rs 29,779.78/g** (34,000 × 87.50 ÷ 99.90), always returned with an `is_derived` flag — never used silently.

Task 3 — The movement ledger

> **FR-14, FR-15, FR-16, FR-21**
> *"One record shape for every gram that moves, whoever it moves to or from… A mistake from last week must be correctable… nothing already recorded can quietly change afterwards… A party balance has two sides that are never netted into one number: metal owed… and money owed."*

What it does: every time metal moves — sold to a customer, bought back, issued to a goldsmith, bought from a wholesaler — it's written down as one "movement." This tracks what every person owes the shop, or is owed, and keeps a full, permanent history.

Test 1 — a customer sale, then a correction

STEP	DETAIL
Sell	10g 22K chain to Ayesha for Rs 224,000
Her balance after	9.166g pure gold owed to the shop
Correct	weight was actually 10.5g, not 10g
Her balance after correction	9.624g — the old 10g entry is excluded, not added on top

RESULT

Full history preserved: **10.000g → 10.500g**, both rows still in the table forever.

Test 2 — the same record shape, a different party type

Proves the brief's own point: one shape works for a customer, a karigar, a wholesaler or another shop.

FIELD	VALUE
Party	Ustad Rafiq (Karigar) — raw metal issued to him
Weight	1 tola (11.6638038g) of 24K gold
Rate	Rs 33,800/g
Amount	Rs 394,236.57

RESULT

His metal balance: **11.652g pure gold** owed back to the shop — the exact same `recordMovement()` call as the customer sale above, just a different party type and direction, with no special-case code anywhere for "karigar."

Task 5 — Buy-back valuation

> **FR-11, FR-12, FR-13, FR-24a, G-3**
>
> *"Gold coming back over the counter is gold being bought fresh today. It is never a sale being undone... Today's weight is the only weight used... The original sale price and weight cannot reach the valuation at all."*

> ***Done when:*** *"the identical one-tola item brought back the same day values below what the customer paid."*

What it does: when a customer brings jewelry back, the shop re-weighs it today and prices it using today's buying rate — never what the customer originally paid. A small deduction covers wear and impurities.

Test 1 — a straightforward buy-back

INPUT	VALUE
Re-weighed today	20g, 22K
Today's buying rate	Rs 22,000/g
Deduction	2.5% (shop default)

RESULT

20.000g − 2.50% (0.500g) = 19.500g × Rs 22,000.00/g = **Rs 429,000.00**

Test 2 — the brief's exact "Done when" test

STEP	DETAIL
Sold today	1 tola (11.6638038g) 22K to Bilal at the selling rate, Rs 22,400/g
Sale amount	**Rs 261,269.21**
Same item, brought back same afternoon	re-weighed at 1 tola, priced at today's buying rate, Rs 22,000/g, 2.5% deduction
Buy-back amount	**Rs 250,184.00**

RESULT

Rs 250,184.00 < Rs 261,269.21 — **YES**, the buy-back values below what was paid, same day, with no way for the original sale figure to reach the calculation at all.

What's deliberately not built

The brief listed two stretch goals — **karigar reconciliation** and the **zakat / historical reporting basis**. Neither was built this week, by design. The Day 1 design note explains how each one lands on top of what's here without needing anything already built to be reshaped.